Manas Karra

UAE Golden Visa Holder | +971-563761966 | manas.karra@gmail.com
manaskarra.com | linkedin.com/in/manas-karra | github.com/manaskarra

Education

BITS Pilani Dubai, UAE
B.E. Computer Science • CGPA: 8.0/10 *Sep. 2021 – Jul. 2025*

- **Relevant Coursework**: Deep Learning, Machine Learning, NLP, Neural Networks & Fuzzy Logic, Data Structures & Algorithms, Computer Networks & OS
- **Professional Assistant**, Digital Design Lab (Sep. – Dec. 2024): Led Verilog/FPGA lab sessions for second-year students.

Skills

Languages & Frameworks: Python, JavaScript/TypeScript, Java, C++, SQL, React, Next.js, Flask, FastAPI, Gradio
LLMs & Agents: LLMs, Multi-Agent Systems, LangChain, LangGraph, Google ADK, MCP, Embeddings, Weaviate, Pinecone, Langfuse
ML & DL: XGBoost, Scikit-learn, TensorFlow, PyTorch, Hugging Face, OpenCV
Data & Storage: Supabase, PostgreSQL, BigQuery, Pandas, NumPy
Cloud & DevOps: GCP, Azure, Docker, Kubernetes, Git, CI/CD

Experience

Applied AI Engineer Feb. 2025 – Present
Deriv *Dubai, UAE*

- Owned and productionized an affiliate analytics platform (**React**, **Flask**, **Supabase**) over a **13M+** row dataset, adopted by **100+** users across growth, finance, and compliance teams. Cut analysis cycles from weeks to seconds via **LLM**-powered regional intelligence and **XGBoost** churn/growth classifiers.
- Built a **RAG**-based natural language BI agent on **Google's ADK** with **Weaviate** hybrid search over hundreds of business tables, with **Langfuse** tracing and evals across a 68-question benchmark scored via Jaccard table-overlap accuracy.
- Engineered an **ambient** affiliate outreach agent orchestrating SQL triage, LLM reasoning, and churn/growth classifiers across lifecycle states, evaluating **60K+** active affiliates daily to automate previously manual outreach with an active learning loop.
- Built a **LangGraph** multi-agent signal generation pipeline for intraday gold/silver trades with false-positive guardrails and **LLM**-based English–Arabic translation, delivered via Telegram to **100+** VIP affiliates.

QA Analyst & Tester Jun. 2023 – Aug. 2023
Impiger Technologies *Dubai, UAE*

- Executed functional, performance, and exploratory testing for a large-scale streaming application across multiple devices, designing and maintaining **700+** test cases while collaborating with engineering teams to identify critical defects and ensure coverage, release stability, and timely delivery.

Projects

verd | pypi.org/project/verd | *Python, Multi-LLM, MCP, Slack, PyPI* Mar. 2026 – Present

- Built and open-sourced on **PyPI** a multi-LLM debate engine shipped as CLI, **MCP** server for Claude Code/Cursor, and Slack bot; achieved **40.0% F1** on the Martian Code Review Benchmark (50 real PRs) — a **+37%** lift over Claude Opus 4.6 solo.

RagaXonic | github.com/manaskarra/RagaXonic-DL | *Python, TensorFlow, LSTM, Gradio* Sep. 2024 – Jan. 2025

- Built a 3-layer **LSTM** network in TensorFlow generating polyphonic MIDI compositions that fuse Hindustani ragas with Western chord progressions, trained with dropout and batch normalization on Music21-parsed data, and shipped via an interactive **Gradio** UI with temperature scaling and note-count controls.

GradeScale | github.com/manaskarra/GradeScale-NLP | *Python, Flask, Hugging Face, NLP* Feb. 2024 – Jun. 2024

- Built a Flask-based automated essay grading system scoring 7 dimensions (accuracy, organization, grammar, vocabulary, punctuation, relevance, coherence) via a stack of **Hugging Face** models — sentence-transformers (all-MiniLM-L6-v2), T5 grammar correction, and domain-tuned coherence/relevance transformers.

ANPR | github.com/manaskarra/ANPR-CV | *Python, OpenCV, Tesseract* Sep. 2023 – Jan. 2024

- Built an end-to-end number plate recognition pipeline in **OpenCV** (bilateral filtering, Canny edge detection, contour approximation via 4-vertex polygon fitting) feeding extracted regions into **Tesseract** OCR, with timestamped detections logged to a Pandas-backed CSV store.